

Class 08 — Homework

Bong Study Hub — Foundation Batch 2026 · Class 08: Making Decisions: `if` / `elif` / `else`

The First Code That "Thinks" Before Acting

Estimated time: 30–45 minutes **Total:** 30 marks · **Optional Challenge:** up to 3 bonus marks **Tools needed:** pen and paper, or a simple document. A computer with Python is helpful for checking your work, but not required to answer any question. **Prerequisite:** Classes 05–08. No nested `if` statements, loops, or `input()` are needed anywhere in this assignment.

What This Assignment Is About

In class, we answered the question Class 07 left open: how do we make a program actually behave differently depending on what's true? We met **conditionals** — `if`, `elif`, and `else` — and learned that **indentation is real Python syntax**, not decoration. We also learned the class's trickiest lesson: in an `elif` chain, **only the first true branch runs**, so order matters.

This assignment asks you to **trace code carefully, not just recall facts about it**. Several questions give you a conditional and ask exactly which branch runs — the `elif`-ordering lesson in particular rewards patient, step-by-step reading rather than guessing.

Instructions

- **Answer in your own words** where a question asks for an explanation.
 - For "predict the output" or "trace the code" questions, write the **exact** output Python would produce.
 - **No nested `if`, loops, or `input()` appear anywhere in this homework** — if your answer to any question involves one of those, simplify it back down to a flat `if`/`elif`/`else` chain.
 - You may refer to your **Student Notes** while answering.
 - If you're unsure, **explain your reasoning anyway** rather than leaving a blank — partial thinking is worth more than an empty space.
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Section A — Core Concepts

Question 1 — What Is a Conditional? (2 marks)

In your own words, **explain what a conditional is**, and what a **condition** must evaluate to for an `if` block to run.

Question 2 — Why Indentation Matters (2 marks)

Explain why indentation matters so much in Python, and what happens if a line that should be indented isn't.

Question 3 — else (2 marks)

Explain when an `else` block runs. Does `else` ever take its own condition?

Question 4 — elif (2 marks)

What does `elif` stand for, and when is its condition actually checked?

Question 5 — Why Order Matters (2 marks)

In your own words, explain why the order of conditions in an `if/elif` chain matters, even when more than one condition is true.

Section B — Reading and Reasoning About Code

Question 6 — Predict the Output (3 marks)

```
temperature = 15

if temperature > 30:
    print("It's hot!")
else:
    print("It's not hot.")
```

Write the exact output.

Question 7 — Predict the Output (3 marks)

```
grade = 72

if grade >= 90:
    print("A")
elif grade >= 80:
    print("B")
elif grade >= 70:
    print("C")
else:
    print("D")
```

Write the exact output, and briefly explain which condition was the first one to be `True`.

Question 8 — Order Matters (3 marks)

```
points = 100

if points >= 50:
    print("Level 1 complete")
elif points >= 100:
    print("Level 2 complete")
```

What does this print? *`points` is `100`, which makes *both*

conditions `True` — explain why only one message actually appears, and which one.

Question 9 — Find the Bug (3 marks)

```
is_weekend = True

if is_weekend:
    print("Sleep in!")
    print("Relax today.")
```

What error would this produce, and why? Rewrite the code correctly.

Question 10 — Comparisons and Logic Inside a Condition (3 marks)

```
temperature = 28
is_sunny = True

if temperature >= 25 and is_sunny:
    print("Perfect beach day!")
else:
    print("Maybe stay in.")
```

Write the exact output, and explain why.

Section C — Class Connection & Your Own Program

Question 11 — Connecting Class 01 and Class 08 (2 marks)

Fill in the blank below, using ideas from today's class:

```
ALGORITHM (Class 01 – imagined as a straight line)
↓
_____ (Class 08)
```

Briefly explain, in one sentence, **how today's class widens the way we picture an algorithm.**

Question 12 — Write Your Own Conditional Chain (3 marks)

Write a short program (on paper is fine) that sorts a temperature into at least **three** categories of your choice — for example, cold, mild, and hot. Use an `if`/`elif`/`else` chain, and make sure your conditions are ordered correctly.

Optional Challenge — Bonus (up to 3 marks)

This section is entirely optional and will not affect your base grade.

Trace the program below very carefully.

```
score = 85
bonus = True

if score >= 60:
    result = "Pass"
elif score >= 80 and bonus:
    result = "Pass with Bonus"
elif score >= 90:
    result = "Distinction"
else:
    result = "Fail"

print(result)
```

What does this print? Then explain: **what would need to change about the order of these conditions** for "Pass with Bonus" to actually be reachable for a score like `85` with `bonus = True`?

Submission Format

Choose whichever is easiest for you:

- **Option A:** Handwrite your answers, then photograph or scan them.
- **Option B:** Type your answers in a simple document.
- **Option C:** Submit as a simple PDF.

Please label each answer with its question number.

Before You Submit — Self-Check

- ☐ I answered all 12 questions.
- ☐ For every "predict the output" or "trace the code" question, I wrote the exact output.
- ☐ Where unsure, I explained my reasoning instead of leaving a blank.
- ☐ I did not use nested `if` statements, loops, or `input()` anywhere.
- ☐ I reviewed my work before submitting.